

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

November 2013

CL306 Transportation Engineering-II

Date: 03.12.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Write classification of Indian Railways. [02]
(b) Write types of gauge with dimensions. Also, write factors affecting choice of gauge in railways. [05]
- Q - 2 (a) Explain in brief plate laying and laying of ballast on track. [04]
(b) Explain turn out with components and labeled sketch. [05]
(c) What is yard? Write types of yards and equipments required in station yards. [05]

OR

- Q - 2 (a) Discuss about various fittings and fixtures used in railway track construction. [04]
(b) Enlist types of station. Write about block-A station with sketch. [05]
(c) Explain the term: slip, gathering line, and crossover. [05]
- Q - 3 (a) What do you understand by geometric design of a railway track? Discuss in brief the parameters which affect it. [05]
(b) Write about various curves used in railways. [05]
(c) Write objectives of signaling. [04]

OR

- Q - 3 (a) What is cant? Explain cant excess and cant deficiency with their values for railway track. [05]
(b) Write types of gradients. What is grade compensation of curves? [05]
(c) Explain mechanical interlocking. [04]

SECTION - II

- Q - 4 (a) How to decide the runway orientation. [03]
(b) Define: (Any Four)
(i) Littoral drift (ii) Artificial harbor (iii) Erosion (iv) Silting (v) fender (vi) cross wind [04]
- Q - 5 (a) Discuss criteria for the site selection for an air port. [05]

- (b) Define current. Explain phenomenon of waves. [05]
(c) Justify: Removal of obstruction nearby airport. [04]

OR

- Q - 5 (a) Give full name of: ICAO, AAI, IAAI, FAA, VOR [05]
(b) Enlist classification of harbour. [05]
(c) Write a note on AAI. [04]

- Q - 6 (a) Determine the radius for a taxiway for a supersonic aircraft to negotiate the curve at a turning speed of 60 kmph. The wheel base is 30 m and the wheel tread is 7.2 m. the airport is of B type as per ICAO [04]

- (b) An airport site at sea level with standard atmospheric conditions, the runway lengths required for take off and landing are 2000 m and 2400 m respectively. The proposed airport situated at an altitude of 150 m. if the airport reference temperature is 20°C and if the effective runway gradient is 0.5 percent, calculate the length of runway to be provided. [07]

OR

- (b) Calculate the actual length of the runway from the following data: [07]

Airport elevation : R. L. 100

Airport reference temperature : 28°C

Basic runway length: 600 m

Highest point along the length : R. L. 98

Lowest point along the length : R. L. 95

- (c) What is wind rose diagram? Give classification of it. [03]
